

**PYTHON
PROGRAMMING
INTERNSHIP REPORT
AT CODEALPHA
10/05/2026 TO 10/06/2026**

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ABSTRACT:

This internship report presents the work completed during the Python Programming Internship. The internship focused on developing practical programming skills using Python and understanding real-world software development techniques.

During the internship, three major projects were completed:

1. Hangman Game
2. Stock Portfolio Tracker
3. Basic Chatbot

These projects were designed to improve logical thinking, problem-solving abilities, file handling concepts, object-oriented programming understanding, and user interaction techniques in Python.

TABLE OF CONTENTS:

- Introduction
- About Python Programming
- Applications of Python
- Internship Tasks Overview
- Task 1: Hangman Game
- Task 2: Stock Portfolio Tracker
- Task 3: Basic Chatbot
- Skills Learned During Internship
- Challenges Faced
- Conclusion
- References

INTRODUCTION:

Python is one of the most popular and powerful programming languages in the world. It is widely used in web development, software engineering, automation, artificial intelligence, machine learning, cybersecurity, and data science.

The purpose of this internship was to gain practical experience in Python programming by developing real-world projects. The internship helped in understanding programming logic, debugging techniques, and project development methodologies.

Through this internship, different Python concepts were implemented practically, improving both theoretical and technical knowledge.



ABOUT PYTHON PROGRAMMING:

Python is a high-level, interpreted, and object-oriented programming language developed by Guido van Rossum in 1991.

Python is known for its:

- Simple syntax
- Readability
- Flexibility
- Cross-platform support
- Large library ecosystem

Python supports multiple programming paradigms including:

- Procedural Programming
- Object-Oriented Programming
- Functional Programming



APPLICATIONS IN PYTHON:

Python is widely used in many fields including:

- Web Development
- Data Science
- Artificial Intelligence
- Machine Learning
- Automation
- Game Development
- Desktop Applications
- Cybersecurity
- Data Analysis

INTERNSHIP TASKS OVERVIEW:

During the internship, three Python projects were developed:

Task No	Project Name
1	Hangman Game
2	Stock Portfolio Tracker
3	Basic Chatbot

These tasks were designed to improve programming concepts and practical implementation skills.



TASK 1: HANGMAN GAME:



(1) INTRODUCTION:

The Hangman Game is a word guessing game developed using Python. In this game, the user attempts to guess a hidden word letter by letter within a limited number of attempts.

The project was developed to improve understanding of loops, conditional statements, string operations, and user interaction.

Start



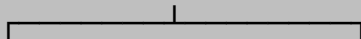
Select Random Word



Take User Input



Correct Guess?



Yes

No



Update Word Reduce Lives



Word Complete? Lives = 0?



Win

Game Over

(2) WORKING PRINCIPLE:

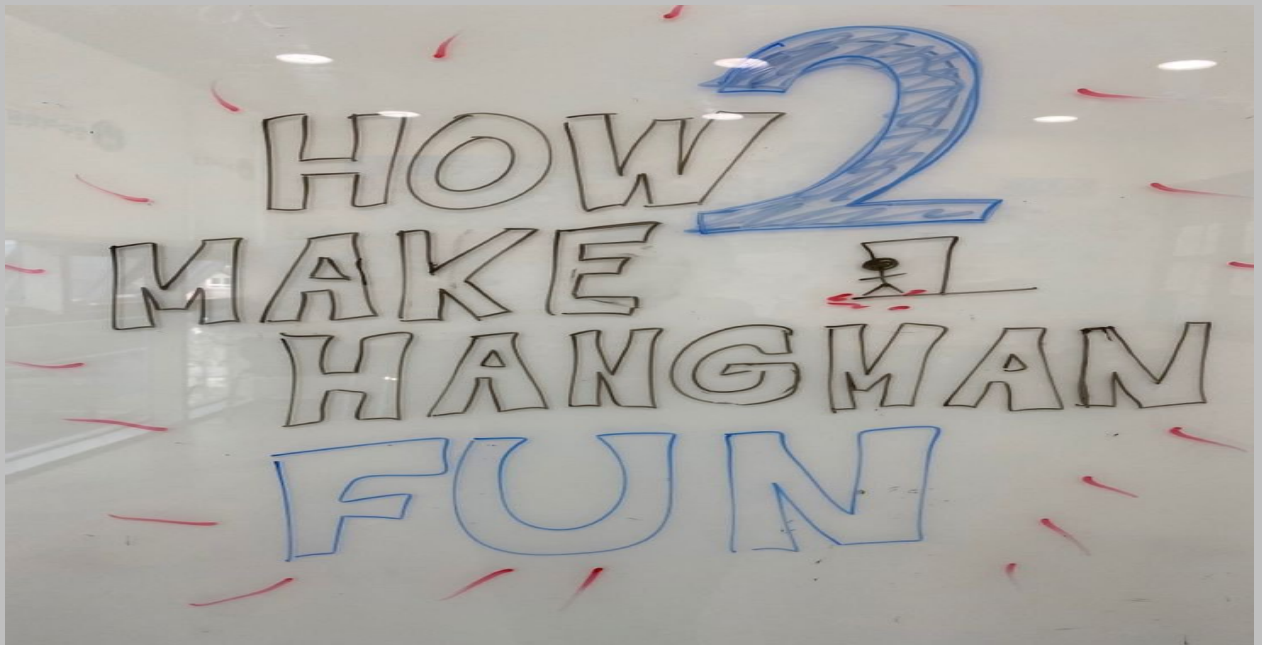
The program randomly selects a word from a predefined list. The user guesses letters one by one. If the guessed letter exists in the word, it is displayed in the correct position. Otherwise, the number of remaining attempts decreases.

The game continues until:

- The user guesses the complete word
OR
- The user runs out of attempts

(3) ALGORITHM:

- Import random module
- Create a list of words
- Select a random word
- Display blank spaces
- Take user input
- Check guessed letter
- Update display word
- Reduce attempts if incorrect
- Repeat until game ends
- Display result



(4) CODE:

```
[1]: import random

words = ["python", "coding", "programming", "computer"]

word = random.choice(words)
guessed = []
attempts = 6

display = ["_"] * len(word)

while attempts > 0 and "_" in display:
    print("\nWord:", " ".join(display))
    guess = input("Enter a letter: ").lower()

    if guess in word:
        for i in range(len(word)):
            if word[i] == guess:
                display[i] = guess
    else:
        attempts -= 1
        print("Wrong guess!")

    print("Remaining attempts:", attempts)

if "_" not in display:
    print("Congratulations! You won.")
else:
    print("Game Over! The word was:", word)
```

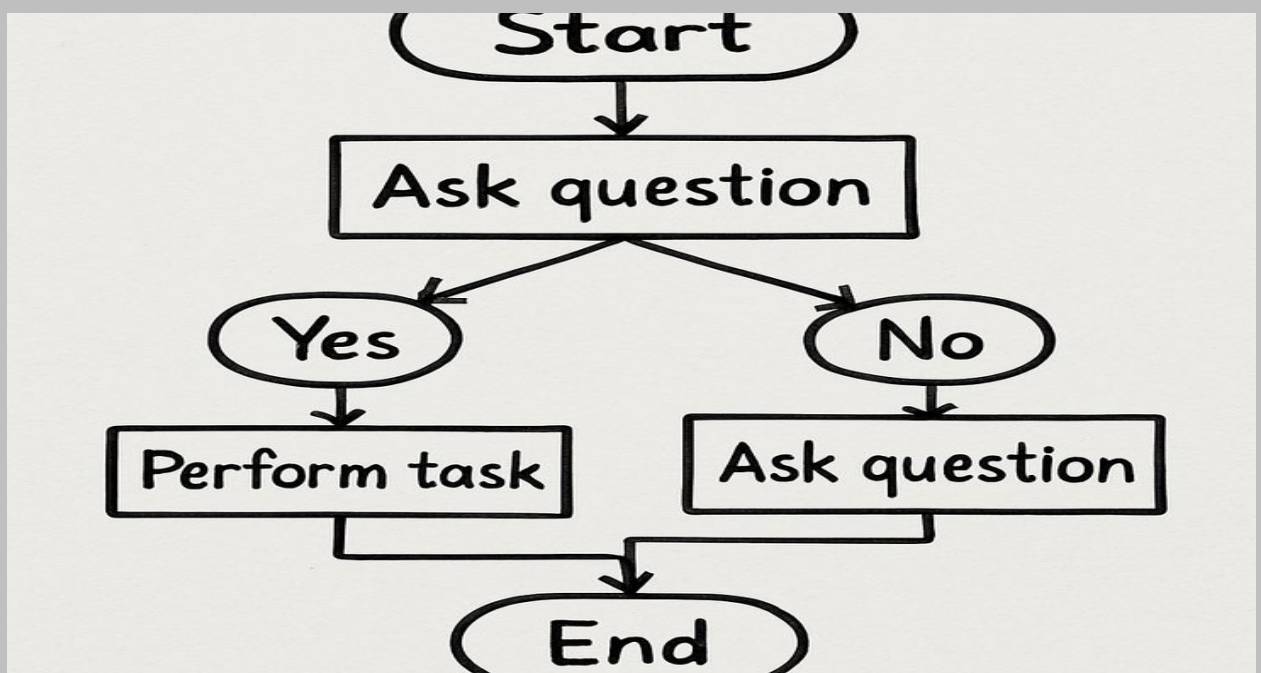
(5) OUTPUT:

The program displays hidden letters using underscores. Users enter guesses through keyboard input. Correct guesses reveal letters while incorrect guesses decrease attempts.

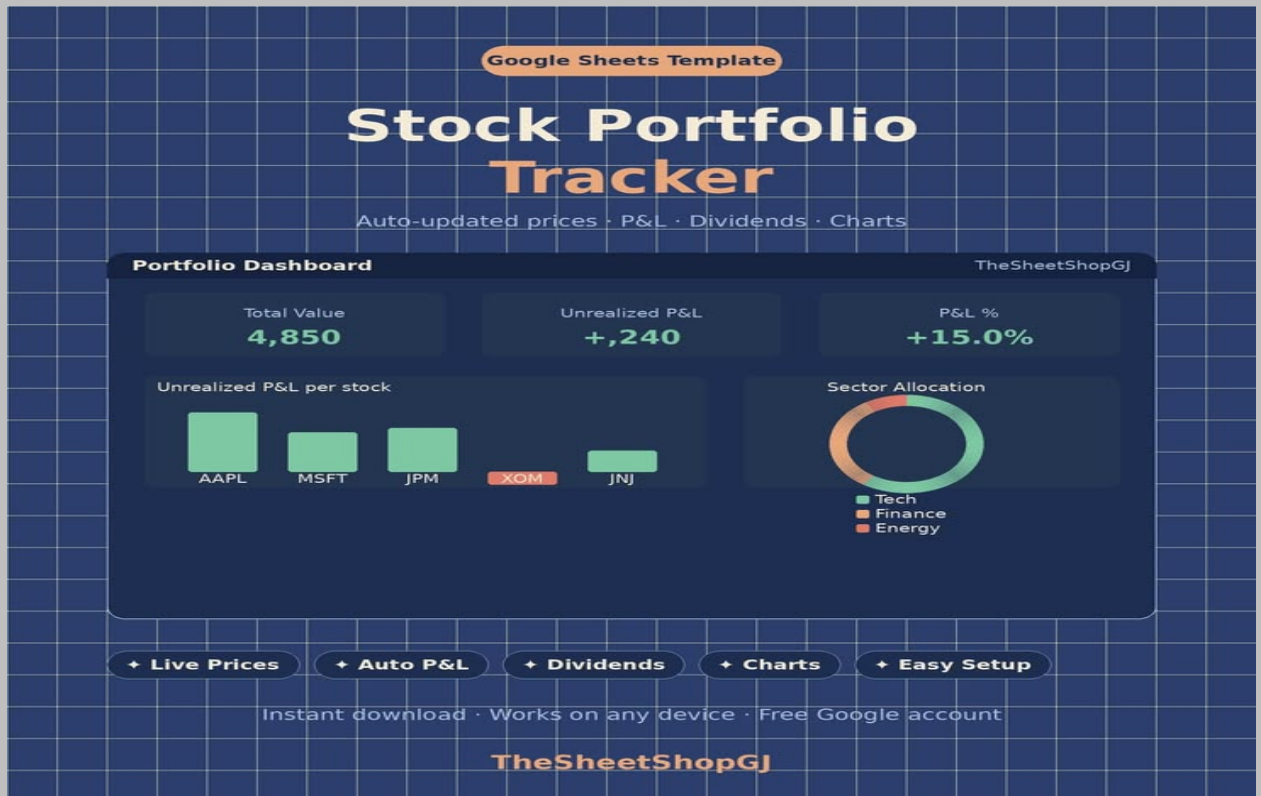
```
Word: _ _ _ _ _  
Enter a letter: c  
Remaining attempts: 6  
  
Word: c _ _ _ _  
Enter a letter: o  
Remaining attempts: 6  
  
Word: c o _ _ _  
Enter a letter: m  
Remaining attempts: 6  
  
Word: c o m _ _  
Enter a letter: p  
Remaining attempts: 6  
  
Word: c o m p _ _  
Enter a letter: u  
Remaining attempts: 6  
  
Word: c o m p u _ _  
Enter a letter: t  
Remaining attempts: 6  
  
Word: c o m p u t _  
Enter a letter: e  
Remaining attempts: 6  
  
Word: c o m p u t e _  
Enter a letter: r  
Remaining attempts: 6  
Congratulations! You won.
```

(6) CONCLUSION:

The Hangman Game project improved logical thinking and strengthened understanding of basic Python concepts. It also enhanced problem-solving abilities and user interaction handling.



TASK 2: STOCK PORTFOLIO TRACKER:



(1) INTRODUCTION:

The Stock Portfolio Tracker is a Python application developed to manage and track stock investments. The program allows users to store stock names, quantities, and prices, then calculates the total investment value.

This project helped in understanding dictionaries, calculations, and data management.

User Input



Stock Data Collection



Portfolio Calculation



Profit/Loss Analysis



Visualization & Report

(2) WORKING PRINCIPLE:

The user enters stock details including:

- Stock Name
- Quantity
- Price

The program calculates:

- Individual stock value
- Total portfolio value

(3) ALGORITHM:

- Create dictionary for stocks
- Take stock details from user
- Store stock information
- Calculate total value
- Display portfolio summary



(4) CODE:

```
[4]: portfolio = {}

n = int(input("Enter number of stocks: "))

for i in range(n):
    stock = input("Enter stock name: ")
    quantity = int(input("Enter quantity: "))
    price = float(input("Enter price: "))

    portfolio[stock] = quantity * price

total = 0

print("\nPortfolio Summary")

for stock, value in portfolio.items():
    print(stock, "=", value)
    total += value

print("Total Investment =", total)
```

(5) OUTPUT:

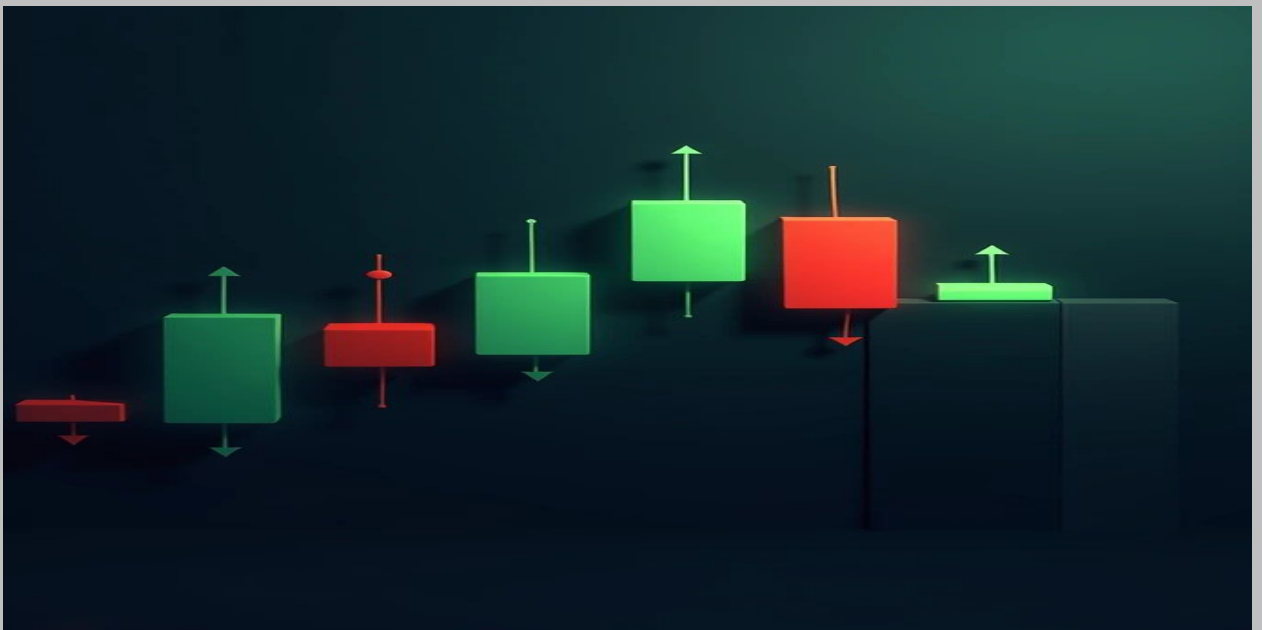
The program displays all stock investments along with the total portfolio value. It helps users track investment amounts efficiently.

```
Enter number of stocks: 2
Enter stock name: TESLA
Enter quantity: 5
Enter price: 300
Enter stock name: HONDA
Enter quantity: 3
Enter price: 100

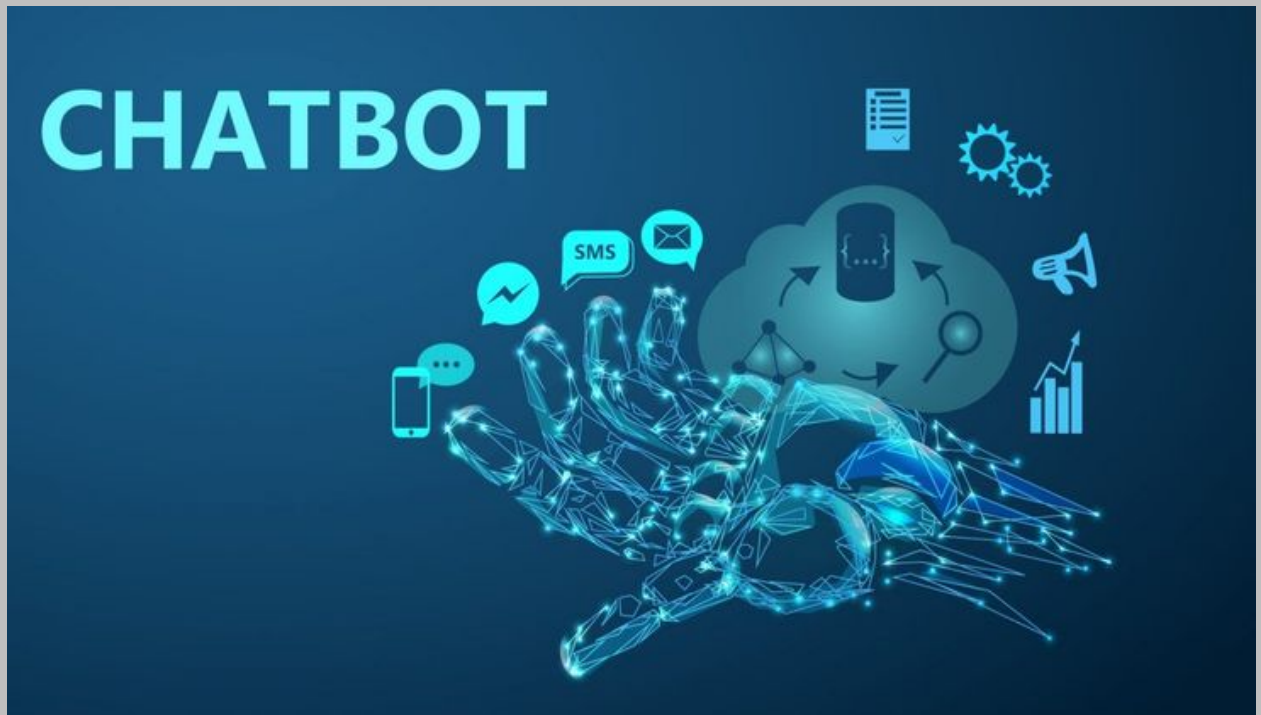
Portfolio Summary
TESLA = 1500.0
HONDA = 300.0
Total Investment = 1800.0
```

(6) CONCLUSION:

The Stock Portfolio Tracker project enhanced programming logic and practical understanding of data handling concepts in Python.



TASK 3: BASIC CHATBOT:



(1) INTRODUCTION:

The Basic Chatbot is a simple conversational application developed using Python. The chatbot interacts with users and responds to predefined questions.

This project introduced basic artificial intelligence concepts and user interaction systems.

User Message



Input Processing



Rule Matching



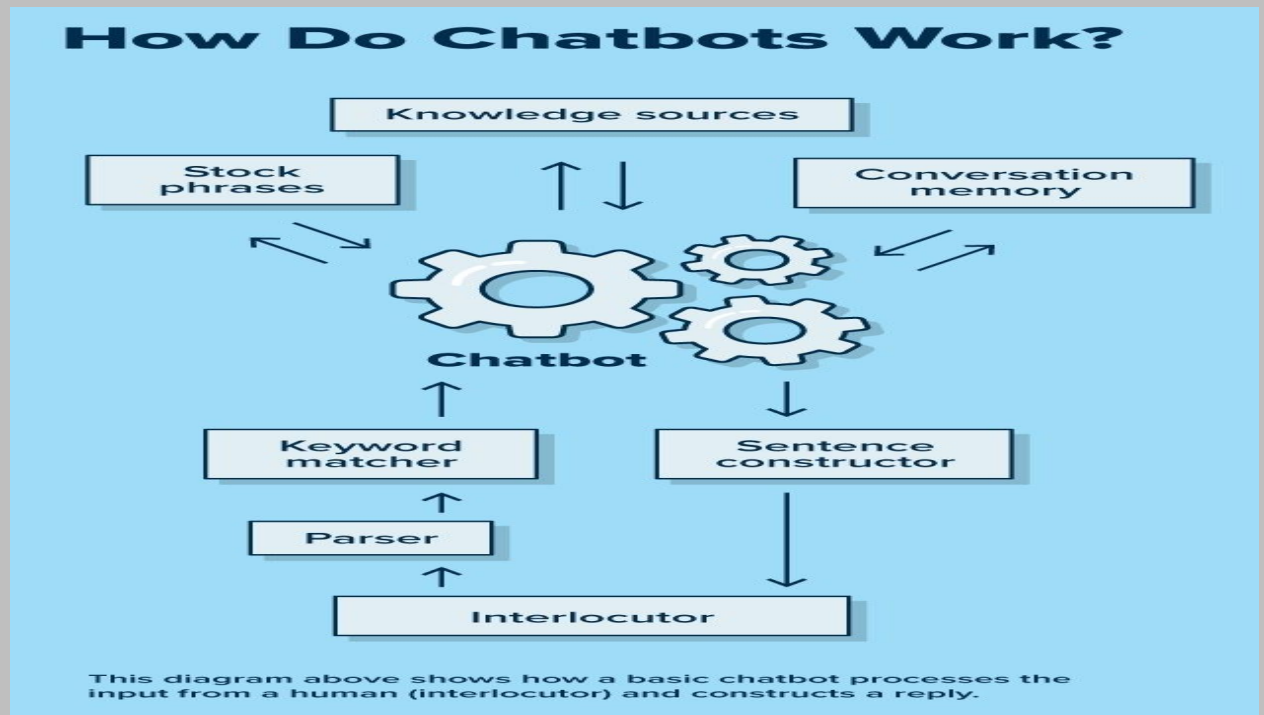
Generate Response



Display Reply

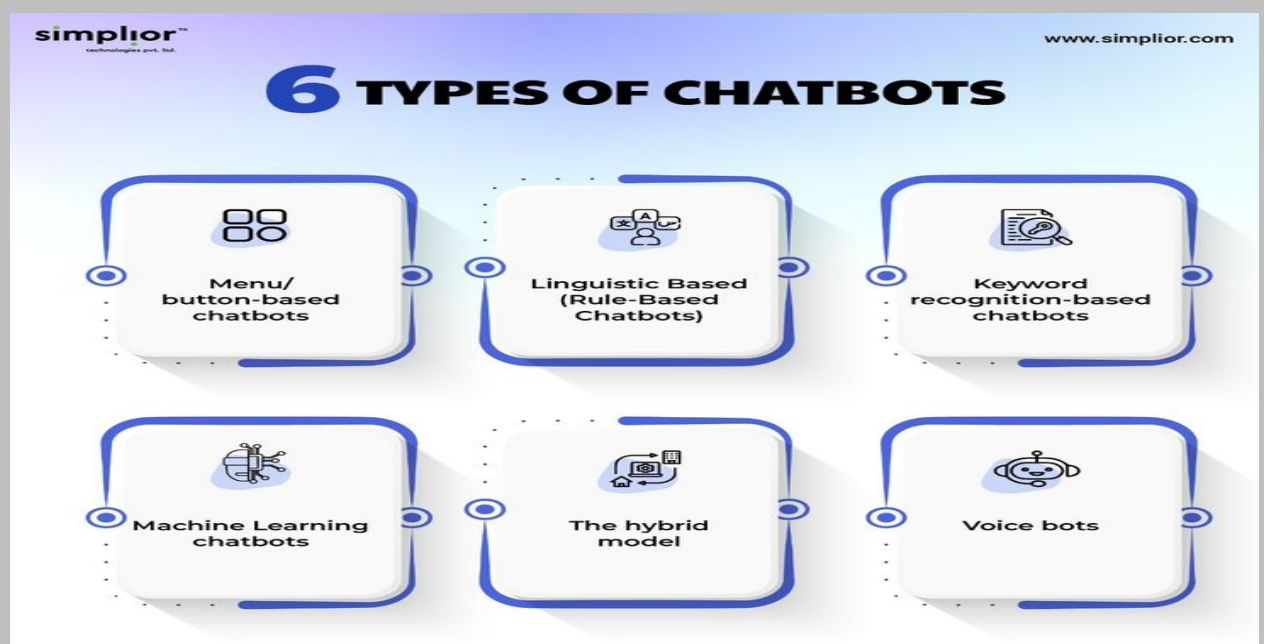
(2) WORKING PRINCIPLE:

The chatbot receives user messages and checks predefined conditions. Based on the input, the chatbot generates appropriate responses.



(3) ALGORITHM:

- Start chatbot
- Take user input
- Convert text to lowercase
- Match conditions
- Display response
- Repeat until exit command



(4) CODE:

```
[ ]: print("Basic Chatbot")
print("Type 'bye' to exit")

while True:
    user = input("You: ").lower()

    if user == "hello":
        print("Bot: Hello! How can I help you?")

    elif user == "how are you":
        print("Bot: I am fine. Thank you!")

    elif user == "your name":
        print("Bot: I am a Python chatbot.")

    elif user == "bye":
        print("Bot: Goodbye!")
        break

    else:
        print("Bot: Sorry, I don't understand.")
```

(5) OUTPUT:

The chatbot responds according to user input. It continues interaction until the user types the exit command.

```
Basic Chatbot
Type 'bye' to exit
You: how are you
Bot: I am fine. Thank you!
You: your name
Bot: I am a Python chatbot.
You: bye
Bot: Goodbye!
```

(6) CONCLUSION:

The Basic Chatbot project improved understanding of conversational systems and Python logic implementation.

Skills Learned During Internship

During the internship, the following skills were developed:

- Python Programming
- Debugging
- Problem Solving
- Logical Thinking
- Console Application Development
- User Interaction Handling
- Data Management
- Documentation Skills

Challenges Faced

Several challenges were faced during the internship including:

- Understanding programming logic
- Debugging errors
- Managing user inputs
- Handling program flow
- Implementing conditions correctly

These challenges improved technical and analytical skills.

Conclusion

The Python Programming Internship was highly informative and beneficial. It provided practical experience in Python development and strengthened programming concepts.

The projects developed during the internship enhanced logical thinking, coding abilities, and software development understanding. The internship also improved confidence in developing real-world Python applications.

Overall, the internship was a valuable learning experience that contributed significantly to technical and professional growth.

References

- Template: CANVA
- Pictures: PINTEREST
- Programming: PYTHON
- Environment: JUPYTER NOTEBOOK

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END OF REPORT

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